

Roll No.

TBT-703(a)

**B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

MARINE BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Elaborate on some of the economically important finfishes in India.

10 Marks (CO1)

OR

- (b) Discuss some of the economically important oysters in India.

10 Marks (CO1)

2. (a) Define Marine Biotechnology and discuss its various scope.

10 Marks (CO1)

OR

- (b) Elaborate on the different applications of Marine Biotechnology.

10 Marks (CO1)

3. (a) Write notes on the varied microbial diversity in a marine ecosystem.

10 Marks (CO1/CO2)

P. T. O.

(2)

OR

(b) What are extremophiles ? Discuss in detail the different kinds of extremophiles. 10 Marks (CO1/CO2)

4. (a) What is a non-culturable microbe ? Discuss, how can metagenomics be studied to identify such microbes. 10 Marks (CO2)

OR

(b) Elaborate on the different roles microbes play in a marine ecosystem. 10 Marks (CO2)

5. (a) Discuss the protocol for collection, isolation, screening and identification of bacteria in marine samples. 10 Marks (CO2)

OR

(b) Discuss the protocol for collection, isolation, screening and identification of fungi in marine samples. 10 Marks (CO2)